

WORLDS OF TOMORROW

AUGUST • 1963 • 50c

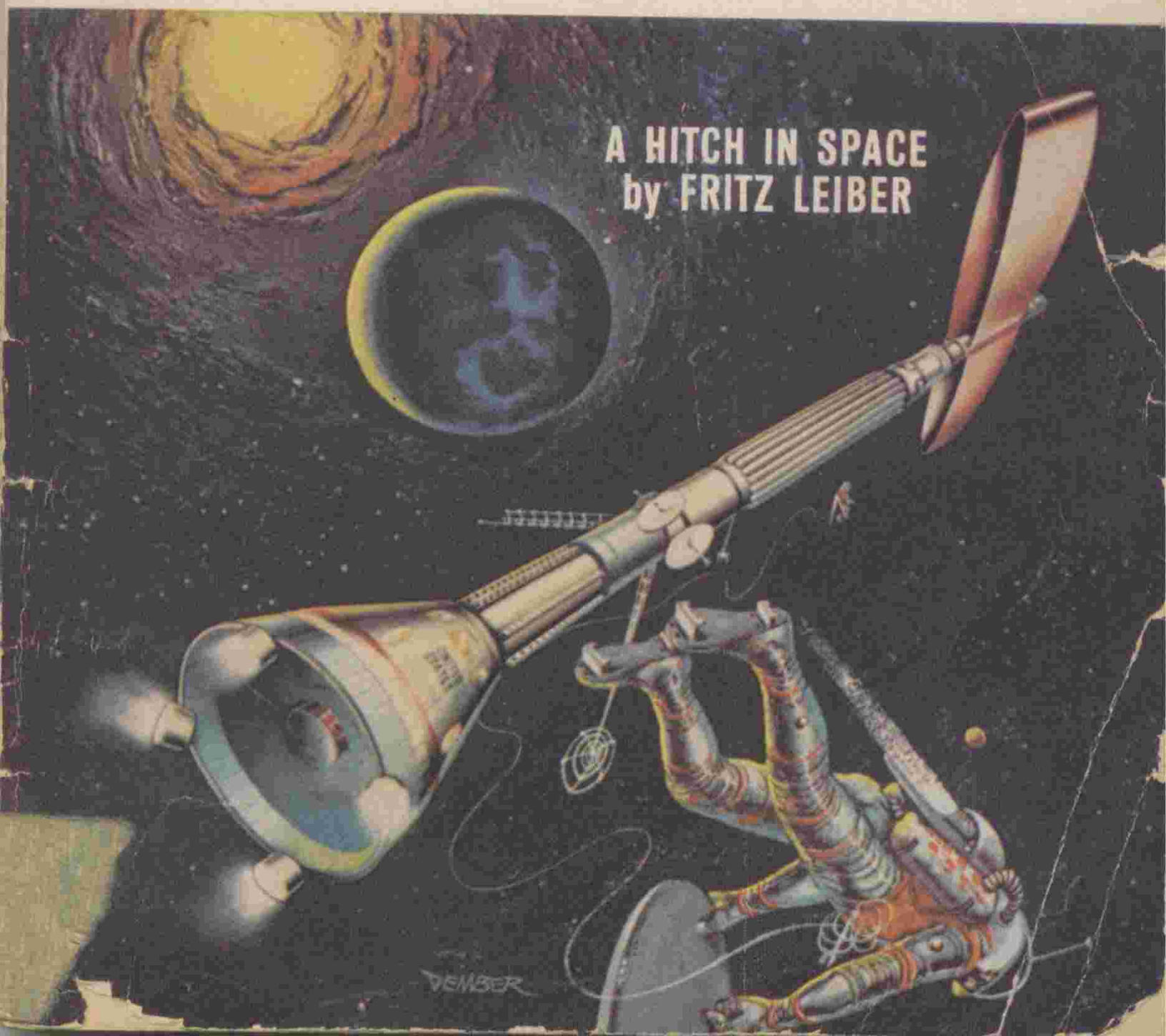
TO THE STARS
J. T. McIntosh

THE IMPOSSIBLE
STAR
Brian W. Aldiss

ALL WE
MARSMEN
Philip K. Dick

THE NEW SCIENCE
OF SPACE SPEECH
Vincent H. Gaddis

A HITCH IN SPACE
by FRITZ LEIBER



WORLDS OF TOMORROW

AUGUST • 1963 • 50c

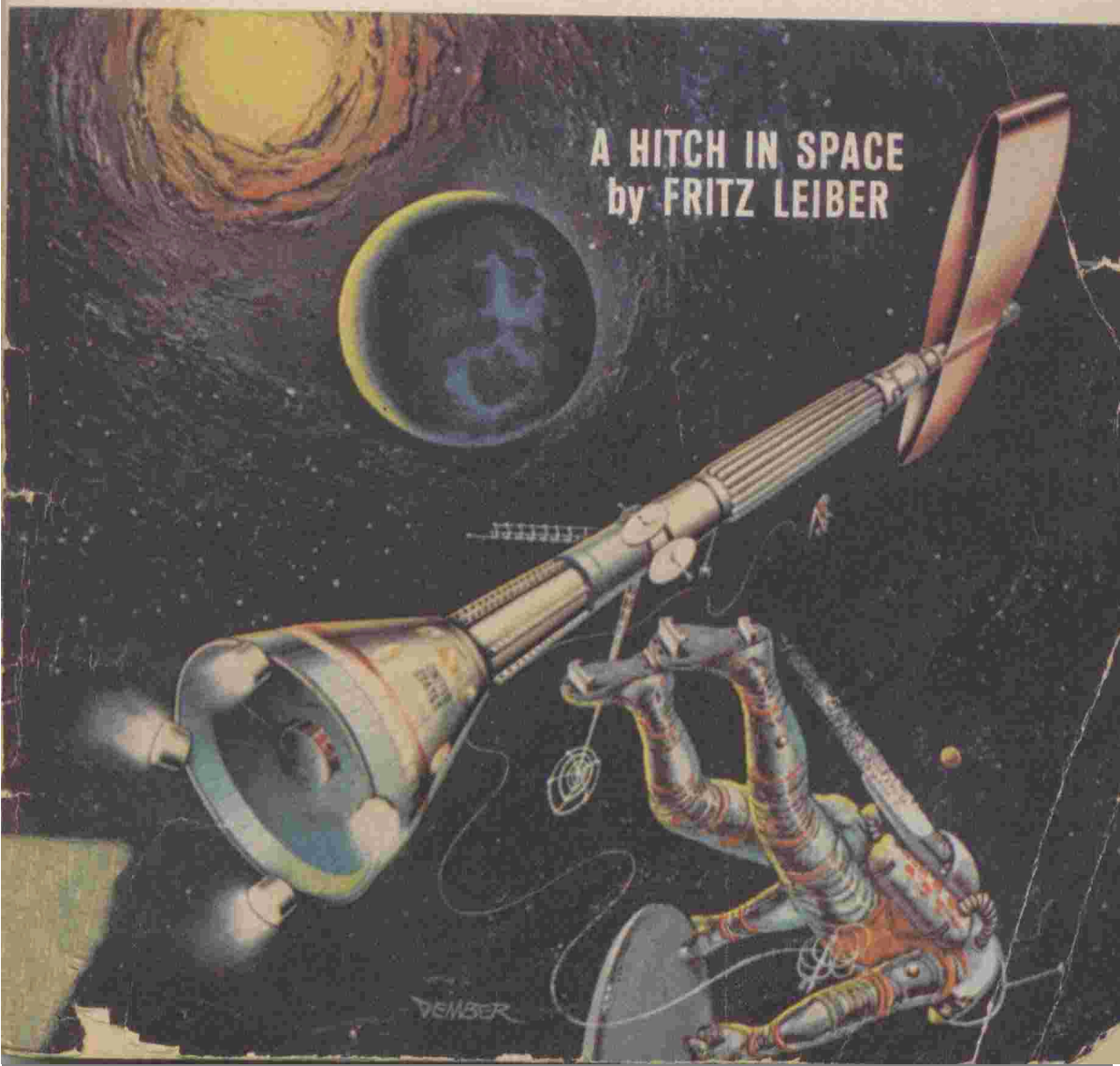
TO THE STARS
J. T. McIntosh

THE IMPOSSIBLE
STAR
Brian W. Aldiss

ALL WE
MARSMEN
Philip K. Dick

THE NEW SCIENCE
OF SPACE SPEECH
Vincent H. Gaddis

A HITCH IN SPACE
by **FRITZ LEIBER**



The Project Gutenberg eBook of The new science of space speech

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The new science of space speech

Author: Vincent H. Gaddis

Release date: August 8, 2025 [eBook #76654]

Language: English

Original publication: New York, NY: Galaxy Publishing Corporation, 1963

Other information and formats: www.gutenberg.org/ebooks/76654

Credits: Roger Frank and Sue Clark

***** START OF THE PROJECT GUTENBERG EBOOK THE NEW
SCIENCE OF SPACE SPEECH *****

The New Science of Space Speech

By Vincent H. Gaddis

How to talk to Martians, dolphins and creatures from the farthest stars—not tomorrow, but now!

A giant ear to listen to the whispers from infinity is being built at Sugar Grove, W. Va. This 600-foot radio telescope, largest ever designed, will cost \$100 million. When completed, its massive antenna, covering 6½ acres, will be trained on the mighty stellar mainland far beyond our solar system.

Astronomers believe that it will pick up cosmic impulses originating in stars from 60 to 80 light-years distant—seven times farther than America's largest existing radio telescope.

Meanwhile, a scientist in the Virgin Islands talks to a frisky dolphin. And the aquatic mammal replies, imitating the man's words with uncanny accuracy.

And at centers of learning in the United States and abroad scholars patiently work over mathematical charts and word lists, seeking formulas that will solve the problem of space speech.

These diverse activities are unified by a common purpose—to intercept and to interpret a possible message from outer space.

This signal across the vast void of the spaceways from intelligent but alien beings will be, perhaps, the most momentous event in human history. It could come tomorrow, or it may not be received for a century or more.

When it does come, man should be prepared to reply. This means we must devise some new method of communication that will transmit thoughts to non-human alien minds.

In awarding a contract for a space speech project, Dr. Dale W. Jenkins, chief of the National Space Administration's environment biology programs, stressed the great need for this knowledge.

“We have not yet determined whether there are any communications directed at earth from outer space,” he said. “If we do make contact, we will have to work out systems of understanding.”

This understanding is an all-important requisite as man reaches out toward the stars.

Understanding, however, will also have to be applied by man to himself when he joins the community of civilizations beyond.

Once interstellar intercourse is established, it will herald a new era in which man will have to recognize another species or form of life as intellectually his equal or more likely his superior. A recent psychological study of the possible effects of outer space contacts indicates that it will deflate human egoism with far-reaching consequences to his culture.

The problem of space speech is two-fold.

First, there are the techniques to be used in actual physical contact with other world inhabitants; second, the far more complex problem of exchanging concepts through the medium of radio communication.

Suppose you are a space explorer. You have landed on Mars or Venus and for the first time you are meeting intelligent creatures that are the products of a completely different line of evolution.

You possess five relatively well-developed senses. If the beings are not hostile, you must first determine if they have the same senses, only some part of them, or additional senses that man does not have.

For example, they may have a sense similar to extra-sensory perception and communicate with each other through telepathy. If you can exchange thoughts with them, that is fine. If you cannot tune in on their mental wavelengths, you're in trouble.

The sense of smell is practically limited to attractive perfumes and repulsive odors. Taste has the same limitations. Touch has been used for communication between humans, as in teaching the blind and deaf, but it requires physical contact (certainly a risky act when meeting strangers) and is limited to elementary concepts at best.

The only practical senses—of those which we humans possess, at any rate—for direct communication are sight and hearing.

If our Martians or Venusians have these senses—and if their reasoning processes are similar to those of humans—then communication could probably be established in the same manner with which we teach our children.

You could use “sign language.” You could point to your mouth and move your jaws to indicate you thought refreshments should be served. You could point to their head or heads (if they had them) and then at your own head and say “head.” With time and patience, a basis for communication could be established.

But suppose their methods of communication are entirely different. Suppose they use antennae, like ants, or gyrations, like bees.

Dr. Karl von Frisch, the German zoologist, discovered that when a bee locates a rich source of nectar, she returns to the hive and performs a dance. The number of times she turns reveals the distance, and her position in relation to the sun and the hive gives the direction.

This “breakthrough” into subhuman communication required controlled and sustained observation. It will have to be the necessary procedure if man encounters creatures with similar characteristics with his present knowledge.

Von Frisch’s discovery was a one-way avenue of understanding. But if the ants and the bees were much larger and more intelligent, we can assume that a demonstrative style of language could be devised for mutual communication.

To our scientists it is obvious that before our spacemen confront alien beings on a distant planet, we must learn the fundamentals of developing communication with a non-human but intelligent species right here on earth. And this is now in progress with “Project Dolphin.”

Bottle-nosed dolphins are not fish, but aquatic mammals. Often, but inaccurately, called porpoises, they are well known as clever, frolicsome entertainers at marineland exhibits.

Dolphins are by far the most intelligent animals other than man, and their brain power in some respects may even be superior to man’s. The dolphin brain is 40 per cent larger than the human, although smaller in proportion to

body weight, and the cerebral cortex—the layer of gray matter that originates rational thought—is just as complicated.

Dr. John C. Lilly, a neurophysiologist and a noted authority on the mammal, is in charge of the project. The research is principally being conducted at the Communications Research Institute of Charlotte Amalie, located at the U.S. Navy base on St. Thomas, Virgin Islands.

Dr. Lilly is working under a contract awarded in 1962 by the National Aeronautics and Space Administration. The contract is for basic scientific research “on the feasibility and methodology for establishing communications between man and other species.”

Dolphins have a complex vocal language. They talk to each other with sharp, high-pitched whistles and they talk almost continuously. Dr. Lilly has determined that the dolphin distress call is “an undulating sound,” with a rasping noise made periodically for range-finding.

Interpreting the dolphin vocabulary will not be easy since the creature emits heavy breathing sounds and there are other masking noises.

In experiments with ESB (electric stimulation of the brain), Dr. Lilly located the portion of the dolphin brain that created a feeling of pleasure. The dolphin almost immediately learned how to turn on a switch producing the current. For comparison, in similar tests it was found that monkeys required 300 or more tries before they attained their ability.

One day the electrical device broke down. The dolphin, annoyed at losing his pleasurable sensation, began making a series of sounds in imitation of the laboratory equipment. Dr. Lilly made a tape recording of these sounds.

Later the doctor played back the recording and in order to more distinctly hear the sounds he decided to run the tape at one quarter its normal speed.

It was then that Dr. Lilly made an astonishing discovery.

With exaggerated slowness, he listened to his own voice on the tape announcing the footage—“three, two, three”—and the dolphin immediately and clearly repeated the words in high-pitched whistles. Other tape recordings of what had seemed to be an unintelligible series of squawks and quacks, when played at half or quarter speed with the sound volume lowered, confirmed the discovery.

The dolphins were not only distinctly imitating the human words they heard, but were compressing their mimicry as to time. They were talking at

a rate eight times faster than humans.

One dolphin, Dr. Lilly recalls, “mimicked my speaking voice so well that my wife laughed out loud, and he copied her laughter.”

When one of the doctor’s assistants who had a southern drawl talked to one dolphin, the animal’s voice came back in clear imitation ... complete with the southern accent.

The next step—and it’s a big one—is to learn the dolphin language. The high-pitched, high-speed chatter must be broken down into definite meanings.

Dr. Frank D. Drake, director of Project Ozma (the recent attempt to receive possible messages by radio telescope), considers the dolphin language study to be of great importance.

He says the project “needs the skills of the radio astronomer in extracting signals from noise, and then the work of the linguist, and, perhaps, the cryptographer. It could well be, if the dolphin studies are correct, that we have right here on earth another intelligent race that is even more alien than some we might encounter in space.”

Second, there is the problem of interpreting and transmitting information through radio communication.

In April, 1960, Project Ozma was launched. The 85-foot radio telescope of the National Radio Astronomy Observatory at Green Bank, W. Va., was focused by government scientists on two stars in an attempt to pick up artificially produced signals.

The stars were Tau Ceti and Epsilon Eridani, 11 light-years or about 66 trillion miles away. They were chosen because they were similar to our own sun in size and rate of rotation.

The frequency at which natural hydrogen emits radio energy in space is 1,420 megacycles, and thus it is a universal constant. Dr. Drake tuned the receiver on both sides of this band.

Day after day the impulses were transmitted to a pen that traced erratic lines on a moving paper roll. But no repetitive pattern appeared that would indicate deliberate signals.

Early in 1961 it was announced that Project Ozma was being suspended. It is expected to be resumed when the new 600-foot radio telescope is completed.

The failure of Project Ozma to receive a message during a few months in operation is no surprise. In fact, it would not be a surprise if no signals were received during daily operation for a millennium.

There are known to be at least 100 quintillion stars. Focusing on one random star in the hope it has a planet having intelligent life beaming signals in our direction is like trying to find a specific drop of water in the ocean.

When a reporter during Project Ozma asked if there was any word from our remote fellow creatures, one scientist told him to come back in 10,000 years.

Yet certain factors may improve these chances. Advanced beings might periodically check the solar systems nearest them to see if they have company. It is not unreasonable to suppose that there is regular cosmic conversation between greatly developed cultures, and if we could detect a channel we might be able to plug in on the party line.

We can only hope, however, that they are using a method we can detect. Man has only recently emerged from savagery and is only beginning to look beyond his little world. To the cosmic callers, our most advanced equipment might be as primitive as smoke signals are to us.

Again, we might be trying to contact beings so entirely different from us that we would have no common ground upon which to build understanding. They might not even respond as we do to the same stimuli. Their appearance, evolution, structure, environment and thinking processes could even be beyond the limits of our imaginations.

But a signal could come—an impulse from out of the boundless abyss telling us we are not alone. What would be the nature of this message? And how could we reply?

Assuming that our senders are using radio wavelengths and have enough similarity to us for mutual understanding, we would first have to isolate the signals from the hash of natural static.

Next, we would have to “crack the code.” The usual cryptographic techniques, which depend on some basic knowledge of the language and

letter frequencies, would not be adequate. We can only hope that the callers give us some clues.

Scientists expect any messages received will be mathematical in nature, since mathematical principles may be regarded as universal constants.

The message might be a simple numeral progression or the numbers of a constant, such as the wave length of the hydrogen atom or the speed of light.

They might send π , for example, the ratio of the circumference of a circle to its diameter. It's a non-stop number, but we would understand if it was worked out to six or eight decimal places. " π from the sky" would be the story of the ages.

Once we had received this signal for recognition and replied in equally simple terms would come the real problem of interpreting or devising a means for transmitting speed.

Hans Fruedenthal, professor of mathematics at the University of Utrecht (Netherlands), has devised a system he calls Lincos (meaning "Lingua Cosmica" or "Cosmic Language"). It consists of teaching the meaning of certain sounds by using numbers. The numbers would be signified by "dots" or "beeps"; the sounds by radio signals of various frequencies and lengths. To illustrate the method, let us assume that the sound "bloop" stands for "equal." Three dots would be sent, then bloop, then three dots. This would be repeated with other numbers until the listeners associated the sound with equal numbers.

The concept of "less than" would similarly be sent by several dots, another sound (like "tweet"), followed by a greater number of dots. The reverse—like a greater number of dots, another sound, and a lesser number of dots—would signify "greater than." Once these concepts were understood, the operative signs like add, subtract, etc. could be taught. Thus a mathematical vocabulary would be established.

Next would come transmitting the length of our basic time unit. The Fruedenthal system would send, say, a four-second dash, followed by the Lincos sound for "second," then four dots. Using different dash lengths with corresponding dots and the same sound, it is assumed that the

recipients would observe that the length of the dash was proportional to the number of dots.

Time concepts (including universal constants) would lead to teaching units of physical length.

Upon this foundation of mathematics, time and dimensions, Lincos develops an ingenious and extensive language for a detailed description of earth, its inhabitants and our culture.

Lincos, of course, assumes that the listeners are capable of understanding our mathematical concepts and that their reasoning processes are similar to ours. It illustrates one great fundamental difficulty in alien communication: whatever system we use, it has to be devised within the limitations of our one-planet knowledge and experience.

The basic principle of association (that is relating numbers to sounds to teach meanings) can be used in other systems. Some form of association, probably beginning with objects and sounds, will be necessary to teach dolphins a human language.

One other fundamental means of communication is being considered by scientists. This is the use of geometrical designs or symbols which would then evolve into pictures. It would be most practical in interplanetary communication.

A picture, as the Chinese say, is worth a thousand words.

In interstellar communication, geometrical figures could possibly be signified by numbers. Thus the pi ratio would denote a circle, three equal successive numbers an equilateral triangle, four equal numbers a square, and so on.

From this elementary basis, a method of translating sounds into drawings could be developed. This might take the form of having electrical circuits attached to pens or tiny lights respond to various sounds, thus transcribing the pictures to paper or film.

The correct interpretation of whatever messages we receive will be of extreme importance. An error could be disastrous.

We need only recall the difficulties we have had in translating early records of our own species to know that interpreting the signals of otherworld beings may be very difficult. Egyptian hieroglyphics were given

many translations that contradicted each other before the Rosetta Stone was found. In one example, there were 12 different translations.

Should this problem develop, we can only hope that the other-worlders are friendly, tolerant and patient.

Then there is the time factor.

If, during Project Ozma, a signal had been received and a reply sent, it would have been 22 years before we knew whether our answer had been received. A reply to a message from 80 light-years away received by the new radio telescope being built would take 160 years for confirmation.

Living languages are fluid. As new words are coined, others become obsolete. Definitions change with passing years.

King George I of England, upon inspecting Sir Christopher Wren's masterpiece, St. Paul's Cathedral, told the famous architect that his creation was "amusing, awful and artificial." Sir Christopher was delighted with the royal compliments.

Three centuries ago amusing meant amazing, awful meant awe-inspiring, and artificial meant artistic.

With time as dimension in universal communication, we would have to choose our words with care.

The accelerated scientific progress of recent years will doubtless continue, with new ways and means of cosmic communication being developed. Radio astronomy itself is barely three decades old. Revolutionary techniques in interstellar contacts may be just around the corner.

Has radio communication with alien beings already occurred? This is a startling possibility.

On August 22, 1924, the planet Mars approached to within thirty-four and a half million miles from the earth. Radio broadcasting stations were silenced and scientists listened for a possible message from across space.

At the suggestion of the late Dr. David Todd, professor emeritus of astronomy at Amherst College, the U.S. Government through diplomatic channels requested that all countries with high-power transmitters silence

their stations for five minutes every hour from 11:50 p.m. August 21 to 11:50 p.m. August 23.

Station WOR, Newark, N.J., reported receiving a word translated as “Eunza.” Other stations announced receiving strange signals.

Twenty-three years later, in 1947, Gene Darling, an early “ham” operator and General Electric Co. employee in Schenectady, N. Y., said he and an assistant had failed to turn off a test transmitter. “It kept on sending out automatic code signals,” he said, “and fearing criticism, we never told of our mistake.”

But something else happened during this 1924 test period of silence that remains a mystery today.

C. Francis Jenkins, of Washington, D. C., had only recently invented a radio photo message continuous-transmission machine. He was asked by Dr. Todd to take a record of any signals received during the periods of silence.

The recording device was attached to a receiver adjusted to the 6,000 meter wave length. Incoming signals caused flashes of light, which were printed on the film by an instrument passing over its surface from side to side. The film was in a roll, 30 feet long and six inches wide, and it was slowly unwound by clockwork under the instrument and light bulb which responded to transmitted sounds.

When the film was developed, it disclosed a fairly regular arrangement of dots and dashes along one side, but on the other side, at almost evenly spaced intervals, were curiously jumbled groups each taking the form of a man’s face.

Scientists at the radio division of the National Bureau of Standards and military code experts examined the film and admitted it was a freak that they couldn’t explain.

“The film of faces is a permanent record that can be studied,” Dr. Todd said, “and who knows just what these signals may have been?”

There have been other incidents. Marconi, the father of wireless, heard strange signals in 1921. And in 1928 Prof. A. M. Low, famous English scientist, listened to a “mysterious series of dots and dashes.”

Ham radio operators have occasionally reported curious stories. In QST, official organ of the International Amateur Radio Union, July, 1950, issue,

Byron Goodman, assistant technical editor of the magazine, tells of a ham receiving strange signals.

Certain unexplainable “echoes” were heard by scientists in 1927, and again in 1928 and 1934 while they were experimenting with the capabilities of radio. The Danish scientist, Hals, and two Scandinavian experimenters, Størmer and Petersen, received echoes from 280,000 to 2,800,000 miles from the earth.

Dr. Arthur C. Clarke reported that in a series of tests in Holland radio echoes of eight seconds delay (corresponding to a reflector at a distance of 744,000 miles) were obtained repeatedly in 1946.

What is the explanation?

Dr. Ronald N. Bracewell, professor of electrical engineering at Stanford University and co-author with J. L. Pawsey of a standard textbook (*Radio Astronomy*, Oxford University Press, 1955), has a theory. He suggests that some of these echoes may have come from a satellite in orbit around our sun.

If highly advanced beings have achieved space travel, placing a satellite in a solar system would be more practical than beaming radio signals continuously at thousands of stars for thousands of years.

Dr. Bracewell suggests that the experimental broadcasts included trigger signals that caused the satellite to respond with echoes. If the satellite’s reply was repeated by man, the satellite would probably release its store of information.

If man does make contact with a superior alien civilization, what will happen?

Recently the Brookings Institution released a report on this question. The study was made for the National Aeronautics and Space Administration at a cost of \$96,000.

If intelligent life is discovered on other worlds, the report warned, the stability of earth’s civilization will be threatened. It recommended a psychological preparation of human beings prior to the discovery.

“While the discovery of intelligent life in other parts of the universe is not likely in the immediate future,” the report said, “it could nevertheless

happen at any time.”

This is the lesson of history: When a culture is faced with a superior culture, it either disintegrates or is changed drastically.

Japan, when it was opened to the outside world, succeeded in adjusting to the new conditions. The Aztec culture collapsed.

Our beliefs, institutions and culture have been based on the premise that man is the most intelligent of creatures. Would we be able to assume a subordinate role?

Perhaps Dr. Otto Struve, the noted astronomer, was thinking about this when newsmen were interviewing him about Project Ozma. “I’m not so sure we should even answer if we did receive such signals,” he said.

Psychological preparation will certainly be needed.

Dr. Harlow Shapley, the Harvard professor emeritus of astronomy, after allowing for all elements of chance among the known stars, conservatively estimates that there should be a million planets with life-producing elements and conditions.

In all the vastness of space and eons of time, there must be intelligent life in myriad forms seeking other intelligent life for interstellar companionship.

When the signal comes, man will answer.

END

Transcriber’s Note

1. This story appeared in the August 1963 issue of *Worlds of Tomorrow*.
2. The original publication used "earth" to refer to the planet "Earth" consistently and "man" to refer to the collective "Man". Retained.

*** END OF THE PROJECT GUTENBERG EBOOK THE NEW
SCIENCE OF SPACE SPEECH ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual

work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in

the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of

exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b)

alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up

to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.